Electronic equipments contain many hazardous heavy metals such as lead, cadmium that can cause severe soil and groundwater pollution.

E-waste dumping yards and the places nearby are polluted and cause severe health hazard.

22.10 Sewage Management

Untreated sewage or wastewater generated from domestic and industrial process is the leading polluter of water sources in India. Sewage water results in agricultural contamination and environmental degradation.

Sources of Sewage/wastewater

- Domestic purpose or household activities
- Dye and textile industries
- Leather industries
- Sugar and breweries industries
- Paper and pulp industries



Figure 22.4 A view of sewage treatment plant

Sewage/wastewater treatment method

The conventional wastewater treatment methods involve the following steps (a) Pre-screening (b) Aeration (c) Sludge Management and (d) Water Reuse.

Pre-screening: Wastewater generated from domestic and industrial activities is screened to remove soil and solid particulates.

Aeration: Screened wastewater is then pumped to an aeration tank. Here the microbial contaminants are removed by the biological degradation that occurs in the presence of air.

Sedimentation process: In this process, the solid particles in suspension form are allowed to settle. The particles that settle out from the suspension is known as sludge.

Sludge removal: The sludge generated by the degradation process is transferred periodically from the tank for safe disposal.

Disinfection: Chlorination and ultraviolet (UV) radiation of treated water is required to remove any microorganism contamination.

Water recycling: The water will then be supplied for domestic or industrial purposes.

22.11 Solid Waste Management

Solid wastes mainly include municipal wastes, hospital wastes, industrial wastes and e- wastes etc. The solid wastes are dumped in the soil which results in landscape pollution.

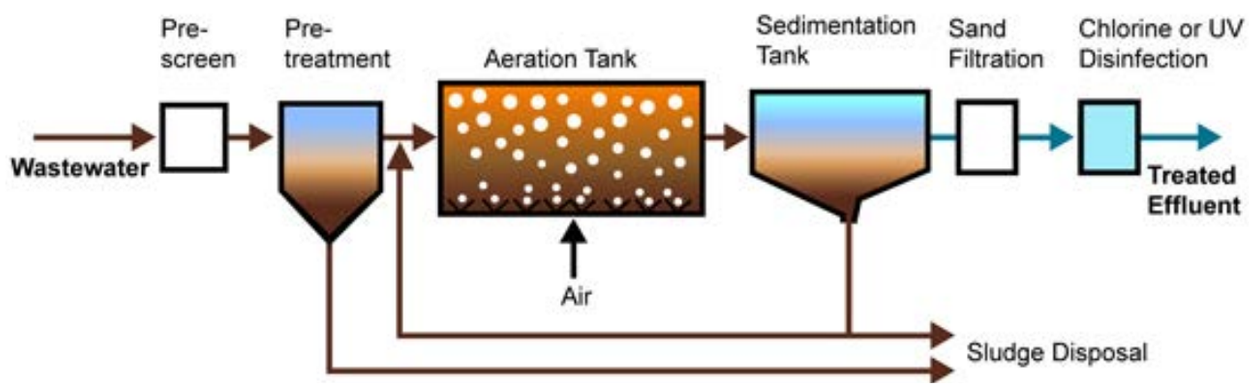


Figure 22.5 Conventional Wastewater Treatment

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